

Measurements (8 th std)

Choose the Appropriate Answer

1. The ratio between the circumference and diameter of any circle is
(a) Area (b) π (c) Radius (d) Perimeter
Ans: (b) π
2. A line segment which joins any two points on a circle is a
(a) Diameter (b) Radius (c) Chord (d) None
Ans: (c) Chord
3. The longest chord of a circle is
(a) Radius (b) Diameter (c) Circumference (d) None
Ans: (b) Diameter
4. The radius of a circle of diameter 24 cm is
(a) 11 cm (b) 12 cm (c) 13 cm (d) None
Ans: (b) 12 Cm
5. A part of circumference of a circle is called as
(a) Diameter (b) Radius (c) An arc (d) None
Ans: (c) an arc
6. The three dimensions of a cuboid are length, breadth and _____.
(a) Width (b) Base (c) Height (d) None
Ans: (c) Height
7. The meeting point of more than two edges in a polyhedron is called as
(a) Vertex (b) Faces (c) Edges (d) None
Ans: (a) Vertex
8. A cube has ----- faces.
(a) 4 (b) 5 (c) 7 (d) 6
Ans: (d) 6
9. The cross section of a solid cylinder is
(a) Square (b) Circle (c) Semi- circle (d) None
Ans: (b) Circle
10. If a net of a 3-D shape has six plane squares, then it is called
(a) Cuboid (b) Sphere (c) Cube (d) None
Ans: (c) Cube
11. The area of a rectangle of length a and breadth b is
(a) $a + b$ (b) $a \cdot b$ (c) $a^2 + b^2$ (d) $2ab$
Ans: (b) $a \cdot b$

12. The area of a square of side a is

- (a) a (b) a^2 (c) $2a$ (d) $4a$

Ans: (b) a^2

13. Write the lateral surface area of a cuboid having length l units, breadth b units and height h units.

- (a) $2(l + b)h$ (b) $2(l + h)b$ (c) $3(l + b)h$ (d) $2(h + b)l$

Ans: (a) $2(l + b)h$

14. The surface area of a cube of edge a is

- (a) $4a^2$ (b) $6a^2$ (c) $3a^2$ (d) a^2

Ans: (b) $6a^2$

15. The total surface area of a cylinder of base radius r and height h is

- (a) $2\pi r(r + h)$ (b) $\pi r(r + h)$ (c) $2\pi r h$ (d) $2\pi r^2$

Ans: (a) $2\pi r(r + h)$

16. The volume of a cuboid of length l , breadth b and height h is

- (a) $l b h$ (b) $l b + b h + h l$ (c) $2(l b + b h + h l)$ (d) $2(l + b)h$

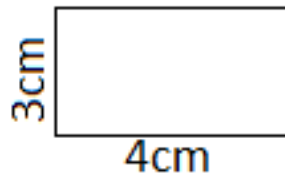
Ans: (a) $l b h$

17. The volume of a cube of edge a is

- (a) a^2 (b) a^3 (c) a^4 (d) $6a^2$

Ans: (b) a^3

18. The perimeter of the figure is



- (a) 7 cm (b) 6 cm (c) 14 cm (d) 16 cm^2

Ans: (c) 14 cm

19. The area of a rhombus is 60 cm^2 . One diagonal is 10 cm. The other diagonal is

- (a) 6 cm (b) 12 cm (c) 3 cm (d) 24 cm

Ans: (b) 12 cm

$$\frac{1}{2} \times 10 \times d_2 = 60 \Rightarrow d_2 = 12 \text{ cm.}$$

20. The area of a trapezium is 40 cm^2 . Its parallel sides are 12 cm and 8 cm. The distance between the parallel sides is

- (a) 1 cm (b) 2 cm (c) 3 cm (d) 4 cm

Ans: (d) 4 cm

$$\frac{(12+8)}{2} d = 40 \Rightarrow d = 4 \text{ cm}$$

21. 8 persons can stay in a cubical room. Each person requires 27 m^3 of air. The side of the cube is

- (a) 6 m (b) 4 m (c) 3 m (d) 2 m

Ans: (a) 6 m

$$\text{Volume} = 8 \times 27 = 216 \text{ m}^3 \Rightarrow \text{side} = \sqrt[3]{216} = 6 \text{ m}$$

22. If the height of a cuboid becomes zero, it will take the shape of a

- (a) Cube (b) Parallelogram (c) Circle (d) Rectangle

Ans: (d) Rectangle

23. The volume of a room is 80 m^3 . The area of the floor is 20 m^2 . The height of the room is

- (a) 1 m (b) 2 m (c) 3 m (d) 4 m

Ans: (d) 4 m

$$V = 80 \text{ m}^3 \Rightarrow \text{area} = 20 \text{ m}^2 \Rightarrow \text{Height} = 80/20 = 4 \text{ m.}$$

24. The floor of a room is a square of side 6 m. Its height is 4 m. The volume of the room is

- (a) 140 m^3 (b) 142 m^3 (c) 144 m^3 (d) 145 m^3

Ans: (c) 144 m^3

$$\text{Volume} = 6 \times 6 \times 4 = 144 \text{ m}^3$$

25. The base radius and height of a right circular cylinder are 14 cm and 5 cm respectively. Its curved surface is

- (a) 220 cm^2 (b) 440 cm^2 (c) 1232 cm^2 (d) $2\pi \times 14 \times (14 + 5) \text{ cm}^2$

Ans: (b) 440 cm^2

$$\text{Curved surface} = 2 \times 22/7 \times 14 \times 5 = 440 \text{ cm}^2$$

26. The heights of the two right circular cylinders are the same. Their volumes are respectively $16\pi \text{ m}^3$ and $81\pi \text{ m}^3$. The ratio of their base radii is

- (a) 16 : 81 (b) 4 : 9 (c) 2 : 3 (d) 9 : 4

Ans: (b) 4 : 9

$$\frac{\pi r_1^2 h}{\pi r_2^2 h} = \frac{16\pi}{81\pi} \Rightarrow \frac{r_1}{r_2} = \frac{4}{9} \Rightarrow 4 : 9$$

27. The ratio of the radii of two right circular cylinders is 1 : 2 and the ratio of their heights is 4 : 1. The ratio of their volumes is

- (a) 1 : 1 (b) 1 : 2 (c) 2 : 1 (d) 4 : 1

Ans: (a) 1 : 1

$$\frac{r_1}{r_2} = \frac{\pi(1)^2 4}{\pi(2)^2 1} = 1 : 1$$

28. A glass in the form of a right circular cylinder is half full of water. Its base radius is 3 cm and height is 8 cm. The volume of water is

- (a) $18\pi \text{ cm}^3$ (b) $36\pi \text{ cm}^3$ (c) $9\pi \text{ cm}^3$ (d) 36 cm^3

Ans: (b) $36\pi \text{ cm}^3$

$$\frac{1}{2} \pi \times 3 \times 3 \times 8 = 36\pi \text{ cm}^3.$$

29. The base area of a right circular cylinder is $16\pi \text{ cm}^2$. Its height is 5 cm. Its curved surface area is

- (a) $40\pi \text{ cm}^2$ (b) $30\pi \text{ cm}^2$ (c) $20\pi \text{ cm}^2$ (d) $10\pi \text{ cm}^2$

Ans: (a) $40\pi \text{ cm}^2$

$$\pi r^2 = 16\pi \Rightarrow r = 4 \text{ cm}$$

$$\text{Curved surface area} = 2 \times \pi \times 4 \times 5 = 40\pi \text{ cm}^2$$

30. The base radius and height of a right circular cylinder are 5 cm and 10 cm. Its total surface area is

- (a) $150\pi \text{ cm}^2$ (b) $300\pi \text{ cm}^2$ (c) 150 cm^2 (d) 300 cm^2

Ans: (a) $150\pi \text{ cm}^2$

$$\text{Total surface area} = 2\pi r (h + r) = 2\pi \times 5 (10 + 5) = 150\pi \text{ cm}^2.$$

31. If the length and breadth of a rectangle are 10 cm and 5 cm, respectively, then its area is:

- (a) 100 sq.cm (b) 150 sq.cm (c) 115 sq.cm (d) 200 sq.cm

Ans: (b) 150 sq.cm

32. The area of a rhombus whose diagonals are of lengths 10 cm and 8.2 cm is:

- (a) 41 cm^2 (b) 82 cm^2 (c) 410 cm^2 (d) 820 cm^2

Ans: (a) 41 cm^2

$$\text{Area of rhombus} = \frac{1}{2} d_1 d_2 = \frac{1}{2} \times 10 \times 8.2 = 41 \text{ cm}^2$$

33. The height of a cuboid whose volume is 275 cm^3 and base area is 25 cm^2 is:

- (a) 10 cm (b) 11 cm (c) 12 cm (d) 13 cm

Ans: (b) 11 cm

$$\text{Height} = \text{Volume} / \text{Base area} = 275/25 = 11 \text{ cm}$$

34. A square is a special case of

- (a) Rhombus (b) Parallelogram (c) Isosceles trapezium (d) Trapezium

Ans: (a) Rhombus

35. One side of an equilateral triangle is 30 cm. Its area is

- (a) $225\sqrt{3} \text{ cm}^2$ (b) 112.5 cm^2 (c) $225\sqrt{2} \text{ cm}^2$ (d) 225 cm^2

Ans: (a) $225\sqrt{3} \text{ cm}^2$

$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} (30)^2 = \frac{\sqrt{3}}{4} \times 900 = 225\sqrt{3} \text{ cm}^2$$

36. Which of the following is the once of a rhombus?

- (a) Product of its diagonals (b) $1/2$ (sum of its diagonals)
 (c) 2 (Product of its diagonals) (d) 2 (Product of its diagonals)

Ans: (c) 2 (Product of its diagonals)

37. If the edge of a cube is 1 cm then which of the following is its volume?

- (a) 6 m^3 (b) 3 m^3 (c) 1 m^3 (d) None of these

Ans: (c) 1 m^3

38. If the parallel sides of a parallelogram are 2 cm apart and their sum is 10 cm then its area is:

- (a) 20 cm^2 (b) 5 cm^2 (c) 10 cm^2 (d) none of these

Ans: (b) 5 cm^2

39. Which of the following has its area and perimeter numerically equal?

- (a) an equilateral triangle of side 1 cm (b) a square of side 1 cm
 (c) a square of side 1 cm (d) a regular pentagon of side 1 cm.

Ans: (c) a square of side 1 cm

40. If the edge of a cube is 1 cm then which of the following is its total surface area?

- (a) 1 cm^2 (b) 4 cm^2 (c) 6 cm^2 (d) None of these

Ans: (c) 6 cm^2

41. Which of the following is equal to 1 kilolitre?

- (a) 1000 millilitres (b) 100 dm^3 (c) 1 dm^3 (d) 1000 dm^3

Ans: (d) 1000 dm^3

42. If the dimensions of a room are l, b and h, ($\therefore l \rightarrow$ length, $b \rightarrow$ breadth and $h \rightarrow$ height) then which of the following is the area of its four walls?

- (a) $2h(1+b)$ (b) $2h(1+h)$ (c) $2l(h+h)$ (d) $2h+1+b$

Ans: (a) $2h(1+b)$

43. If the dimensions of a room are 2 m, 3 and 4 m then which of the following is the number of cubes of size $\frac{1}{2} \times \frac{1}{3} \times \frac{1}{4}$ which can be placed in the room?

- (a) 960 (b) 672 (c) 676 (d) 576

Ans: (d) 576

44. If base area of a room is 12 m^2 and height is 3 m then its volume is:

- (a) 4 m^3 (b) 36 m^3 (c) 12 m^3 (d) 18 m^3

Ans: (b) 36 m^3

45. Two identical cubes each of total surface area of 6 cm^2 are joined end to end. Which of the following is the total surface area of the cuboid so formed?

- (a) 12 cm^2 (b) 18 cm^2 (c) 10 cm^2 (d) 8 cm^2

Ans: (c) 10 cm^2

46. Which of the following quadrilaterals has two pairs of adjacent sides equal and diagonals intersecting at right angles?

- (a) Square (b) Rhombus (c) Kite (d) Rectangle

Ans: (d) Rectangle

47. Which of the following quadrilaterals has a pair of opposite sides parallel?

- (a) Rhombus (b) Trapezium (c) Kite (d) Rectangle

Ans: (b) Trapezium

48. Which of the following quadrilaterals is a regular quadrilateral?

- (a) Rectangle (b) Square (c) Rhombus (d) Kite

Ans: (b) Square

49. Which of the quadrilaterals has all angles as right angles, opposite sides equal and diagonals bisect - each other?

- (a) Rectangle (b) Rhombus (c) Square (d) None of these

Ans: (a) Rectangle

50. Which of the parallelograms has all sides equal and diagonals bisect each other at right angle?

- (a) Square (b) Rectangle (c) Rhombus (d) Trapezium

Ans: (c) Rhombus

51. In an isosceles parallelogram, we have:

- (a) Pair of parallel sides as equal (b) Pair of non-parallel sides as equal
(c) Pair of non-parallel sides as perpendicular (d) None of these.

Ans: (b) Pair of non-parallel sides as equal

52. Which of the following is true for the adjacent angles of a parallelogram?

- (a) They are equal to each other (b) They are complementary angles
(c) They are supplementary angles (d) None of these

Ans: (c) They are supplementary angles

53. The sides of a pentagon are produced in order. Which of the following is the sum of its exterior angles?

- (a) 540° (b) 180° (c) 720° (d) 360°

Ans: (d) 360°

54. Which of the following is a formula to find the sum of interior angles of a quadrilaterals of n-sides?

- (a) $\frac{n}{2} \times 180^\circ$ (b) $\left(\frac{n+1}{2}\right) \times 180^\circ$ (c) $\left(\frac{n-1}{2}\right) \times 180^\circ$ (d) $(n-2) \times 180^\circ$

Ans: (d) $(n-2) \times 180^\circ$

55. Diagonals of which of the following quadrilaterals do not bisect it into two congruent triangles?

- (a) Rhombus (b) Trapezium (c) Square (d) Rectangle.

Ans: (b) Trapezium

56. How many number of faces does a solid sphere has?

- (a) 1 (b) 2 (c) Many (d) None

Ans: (a) 1

57. How many number of vertices does a cone has?

- (a) 1 (b) 2 (c) 3 (d) None of these

Ans: (a) 1

58. How many number of faces does a hemisphere has?

- (a) 1 (b) 2 (c) Many (d) None of these

Ans: (b) 2

59. Which of the following is the number of vertices of sphere?

- (a) 0 (b) 1 (c) 2 (d) 4

Ans: (a) 0

60. If F, E and V represent the faces, edges and vertices respectively of a polyhedral then which of the following is the Euler's formula?

- (a) $F - V + E = 2$ (b) $F + V + E = 2$
(c) $F + V - E = 2$ (d) $F + V = 2 - E$

Ans: (c) $F + V - E = 2$

61. Which of the following is not a quadrilateral?

- (a) Square (b) Rectangle (c) Triangle (d) Parallelogram

Ans: (c) Triangle

62. Which of the following quadrilaterals has two pairs of adjacent sides equal and its diagonals intersect at 90 degrees?

- (a) Square (b) Kite (c) Rhombus (d) Rectangle

Ans: (b) Kite

63. Which one of the following is a regular quadrilateral?

- (a) Square (b) Trapezium (c) Kite (d) Rectangle

Ans: (a) Square

64. If AB and CD are two parallel sides of a parallelogram, then:

- (a) $AB > CD$ (b) $AB < CD$ (c) $AB = CD$ (d) None of the above

Ans: (c) $AB = CD$

65. The perimeter of a parallelogram whose parallel sides have lengths equal to 12 cm and 7 cm is:

- (a) 21 cm (b) 42 cm (c) 19 cm (d) 38 cm

Ans: (d) 38 cm

$\begin{aligned}\text{Perimeter of parallelogram} &= 2 (\text{Sum of Parallel sides}) \\ &= 2 (12 + 7) = 2 (19) = 38 \text{ cm}\end{aligned}$

66. If $\angle A$ and $\angle C$ are two opposite angles of a parallelogram, then:

- (a) $\angle A > \angle C$ (b) $\angle A = \angle C$ (c) $\angle A < \angle C$ (d) None of the above

Ans: (b) $\angle A = \angle C$

Opposite angles of a parallelogram are always equal

67. If $\angle A$ and $\angle B$ are two adjacent angles of a parallelogram. If $\angle A = 70^\circ$, then $\angle B = ?$
(a) 70° (b) 90° (c) 110° (d) 180°

Ans: (c) 110°

The adjacent angles of parallelogram are supplementary.

68. ABCD is a rectangle and AC & BD are its diagonals. If AC = 10 cm, then BD is:
(a) 10 cm (b) 5 cm (c) 15 cm (d) 20 cm

Ans: (a) 10 cm

The diagonals of a rectangle are always equal.

69. Each of the angles of a square is:
(a) Acute angle (b) Right angle (c) Obtuse angle (d) 180 degrees

Ans: (b) Right angle

All the angles of square is at right angle.

70. The quadrilateral whose diagonals are perpendicular to each other is:
(a) Parallelogram (b) Rectangle (c) Trapezium (d) Rhombus

Ans: (d) Rhombus

71. If the two angles of a triangle are 80° and 50° , respectively. Find the measure of the third angle.

(a) 50° (b) 60° (c) 70° (d) 80°

Ans: (a) 50°

Sum of all the angles of a triangle = 180°

$$80^\circ + 50^\circ + x = 180^\circ \Rightarrow x = 180^\circ - 130^\circ = 50^\circ$$

72. In a parallelogram ABCD, angle A and angle B are in the ratio 1:2. Find the $\angle A$.
(a) 30 (b) 45° (c) 60° (d) 90°

Ans: (c) 60°

73. The angles of a quadrilateral are in ratio 1:2:3:4. Which angle has the largest measure?

(a) 120° (b) 144° (c) 98° (d) 36°

Ans: (b) 144°

$x + 2x + 3x + 4x = 360^\circ$ [Angle sum property of quadrilateral]

$$10x = 360^\circ \quad x = 36^\circ$$

74. The length and breadth of a rectangle is 4 cm and 2 cm respectively. Find the perimeter of the rectangle.

- (a) 12 cm (b) 6 cm (c) 8 cm (d) 16 cm

Ans: (a) 12 cm

75. The diagonals of a rectangle are $2x + 1$ and $3x - 1$, respectively. Find the value of x .

- (a) 1 (b) 2 (c) 3 (d) 4

Ans: (b) 2

$$2x + 1 = 3x - 1 \Rightarrow 1 + 1 = 3x - 2x \Rightarrow 2 = x$$

76. The diagonals of a kite:

- (a) Bisects each other (b) Are perpendicular to each other
(c) Does not bisect each other (d) None of the above

Ans: (b) Are perpendicular to each other

77. A rhombus has a side length equal to 5 cm. Find its perimeter.

- (a) 25 (b) 10 (c) 20 (d) 30

Ans: (c) 20

78. ABCD is a parallelogram. If angle A is equal to 45° , then find the measure of its adjacent angle.

- (a) 135° (b) 120° (c) 115° (d) 180°

Ans: (a) 135°

79. A cuboid has _____ pairs of identical faces.

- (a) 2 (b) 3 (c) 4 (d) 5

Ans: (b) 3

80. All six faces of a cube are:

- (a) Identical (b) Different (c) Circular (d) Rectangular

Ans: (a) Identical

81. A cylindrical box has _____ curved surface and _____ circular faces, which are identical.

- (a) One, One (b) One, two (c) two, one (d) two, two

Ans: (b) One, Two

82. A cube has _____ faces.

- (a) 4 (b) 5 (c) 6 (d) 8

Ans: (c) 6

83. Number of vertices a cone has:

- (a) One (b) Two (c) Three (d) Four

Ans: (a) One

84. A sphere has how many faces?

- (a) One (b) Two (c) Four (d) None

Ans: (d) None

85. An example of the cone is:

- (a) Tube (b) A tent (c) A ball (d) A box

Ans: (b) A tent

86. The top view of a cone looks like this:

- (a) A Circle (b) A Square (c) A Rectangle (d) A Triangle

Ans: (a) A Circle

87. The top view of a cube looks like a:

- (a) Circle (b) Square (c) Rectangle (d) Triangle

Ans: (b) Square

88. The top view of a cuboid looks like a:

- (a) Circle (b) Square (c) Rectangle (d) Triangle

Ans: (c) Rectangle

89. The side-view of a cone appears as:

- (a) Circle (b) Square (c) Rectangle (d) Triangle

Ans: (d) Triangle

90. The base of a prism is:

- (a) Circle (b) Triangle (c) Square (d) Any shape

Ans: (d) Any shape

91. An example of a cylindrical shape is:

- (a) Pipes (b) Gas cylinder (c) Cold drink cans (d) All of the above

Ans: (d) All of the above

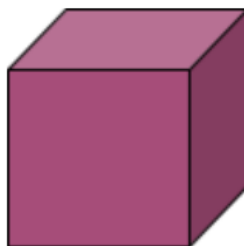
92. What is the name of the shape?



- (a) Circle (b) Square (c) Cone (d) Cylinder

Ans: (a) Circle

93. What is the name of the given shape?



- (a) Square (b) Rectangle (c) Cuboid (d) Cube

Ans: (d) Cube

94. Find the name of the given shape.



- (a) Square (b) Rectangle (c) Cuboid (d) Cube

Ans: (c) Cuboid

95. Prisms and Pyramids are:

- (a) Polygons (b) Cubes (c) Polyhedrons (d) Spherical shapes

Ans: (c) Polyhedrons

96. The perimeter of a rectangle with length (l) and width (w) is:

- (a) $l + w$ (b) $2(l + w)$ (c) $(l + w)^2$ (d) $l w$

Answer: (b) $2(l + w)$

97. If the length and breadth of a rectangle are 15cm and 10cm, respectively, then its area is:

- (a) 100 sq.cm (b) 150 sq.cm (c) 115 sq.cm (d) 200 sq.cm

Ans: (b) 150 sq.cm

98. Volume of a cuboid of length (l), width (w) and height (h) is:

- (a) $l b h$ (b) $l b + b h + h l$ (c) $2(l b + b h + h l)$ (d) $2(l + b) h$

Ans: (a) $l b h$

99. Volume of a cylinder with base radius = r and height = h, is:

- (a) $2\pi r h$ (b) $\pi r^2 h$ (c) $2\pi r (r + h)$ (d) $\frac{1}{3} \pi r^2 h$

Ans: (b) $\pi r^2 h$

100. Surface area of cube of edge 'a' is:

- (a) $4a^2$ (b) $6a^2$ (c) $3a^2$ (d) a^2

Ans: (b) $6a^2$